Data Modelling and Form Design for a Business Application in Zoho Creator

1. Aim

To design a suitable data model and create forms in Zoho Creator for a business application by identifying data fields, defining relationships between forms, and organizing records for efficient data collection and management.

2. Procedure

1. Log in to Zoho Creator: Open Zoho Creator and sign in to your account.

2. Create the application: Open the Rainwater Harvesting Record System application.

3. Identify the data entities: Identify the main entities required for the application, such as rainfall records, storage tanks, and water usage.

4. Design the data model: Decide the fields required for each entity and identify relationships between the entities.

5. Create the Rainfall Record form: Add fields such as Record Date, Location, Rainfall Amount (mm), Collection Area, Estimated Water Collected, and Weather Notes.

6. Create the Storage Tank form: Add fields such as Tank Name, Tank Location, Tank Capacity (Litres), Current Water Level (Litres), Last Inspection Date, and Tank Notes.

7. Create the Water Usage form: Add fields such as Usage Date, Water Quantity (Litres), Usage Purpose, Recorded By, and Usage Notes.

8. Establish relationships: Add a lookup field named Tank Lookup in the Water Usage form and connect it to the Storage Tank form. Configure it to display the Tank Name so that water usage records can be associated with the correct tank.

9. Configure field properties: Set suitable field types, mandatory fields, dropdown choices, and validation rules wherever required.

10. Test the forms: Enter sample records, save them, and verify that the data is stored correctly and the tank lookup displays the available tank names.

3. Data Model

The data model identifies the main entities, their attributes, and the relationship between them.

Rainfall_Record

Record_Date (Date)

Location (Text)

Rainfall_Amount_mm (Number)

Collection_Area (Number)

Estimated (Number)

Weather_Notes (Text)

Storage_Tank

Tank_Name (Text)

Tank_Location (Text)

Tank_Capacity (Number)

Current_Water_Level (Number)

Last_Inspection_Date (Date)

Tank_Notes (Text)

Lookup relationship through Tank_Lookup

Water_Usage

Usage_Date (Date)

Water_Quantity_Litres (Number)

Usage_Purpose (Dropdown)

Tank_Lookup (Lookup to Storage_Tank)

Recorded_By (Text)

Usage_Notes (Text)

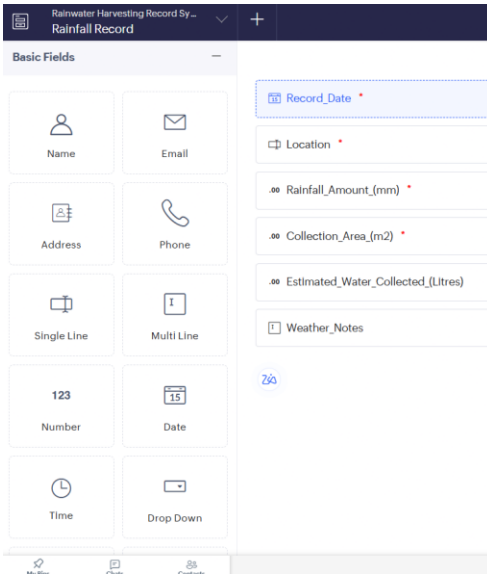
The lookup relationship allows multiple water usage records to reference a storage tank. This helps organize water consumption data by tank.

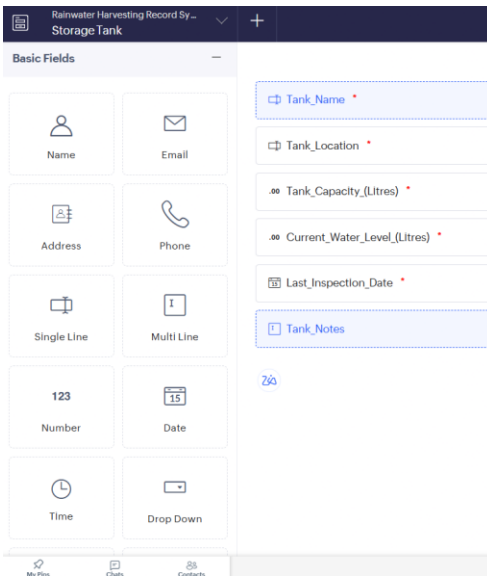
4. Form Design Details

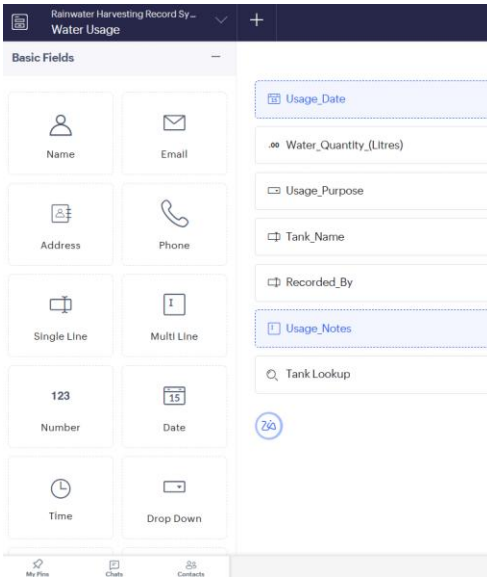
Form	Purpose	Important fields
Rainfall Record	Record rainfall measurements	Date, location, rainfall amount, collection area
Storage Tank	Manage storage capacity and water levels	Tank name, capacity, current water level

Water Usage	Track water consumption	Usage date, quantity, purpose, tank lookup

OUTPUT







5. Result

The data model and forms for the Rainwater Harvesting Record System were designed using Zoho Creator. Three forms—Rainfall Record, Storage Tank, and Water Usage—were organized with suitable fields and data types.

A lookup relationship was configured between the Water Usage and Storage Tank forms, allowing users to select a tank by its name when recording water consumption. The forms support structured data collection and provide a foundation for generating reports, dashboards, and workflow automations.

Thus, the experiment demonstrated how data modelling and form design help organize information and build a functional business application using a low-code development platform.